

# GearDesign Report

Cylindrical gear(pair)

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# 1. Summary

## 1.1. Gear geometry

Meshing pair: Gear1 - Gear2

|                                              | Symbol     | Pair1<br>Gear1                             | Pair1<br>Gear2                             |
|----------------------------------------------|------------|--------------------------------------------|--------------------------------------------|
| Normal module (mm)                           | $m_n$      | 2.0000                                     |                                            |
| Pressure angle at normal section (deg)       | $\alpha_n$ | 20.0000                                    |                                            |
| Helix angle at reference circle (deg)        | $\beta$    | 0.0000                                     |                                            |
| Manual normal backlash (mm)                  | $j_{bn}$   | 0.0000                                     |                                            |
| Number of teeth                              | $z$        | 28.0000                                    | 41.0000                                    |
| Facewidth (mm)                               | $b$        | 45.0000                                    | 45.0000                                    |
| Profile shift coefficient                    | $x$        | 0.3132                                     | 0.2129                                     |
| Quality (ISO)                                | $Q$        | 6.0000                                     | 6.0000                                     |
| Center distance (mm)                         | $a$        | 70.0000                                    |                                            |
| Tooth thickness tolerance                    | -          | DIN 3967 cd25                              | DIN 3967 cd25                              |
| Gear material                                | -          | DIN 18CrNiMo7 ( $\sigma_{Hlim} = 430$ Mpa) | DIN 18CrNiMo7 ( $\sigma_{Hlim} = 430$ Mpa) |
| Reference diameter (mm)                      | $d$        | 56.0000                                    | 82.0000                                    |
| Tip diameter (mm)                            | $d_a$      | 61.253 (61.253 / 61.153)                   | 86.851 (86.851 / 86.751)                   |
| Root diameter (mm)                           | $d_f$      | 52.253 (52.060 / 51.951)                   | 77.851 (77.659 / 77.549)                   |
| Number of teeth spanned                      | $k$        | 4.0000                                     | 5.0000                                     |
| Base tangent length (no backlash) (mm)       | $W_k$      | 21.878 (21.812 / 21.774)                   | 28.009 (27.943 / 27.905)                   |
| Effective Diameter of ball/pin (mm)          | $D_M$      | 3.7500                                     | 3.5000                                     |
| Diametral measurement over two balls<br>(mm) | $M_{dk}$   | 62.801 (62.655 / 62.571)                   | 87.680 (87.516 / 87.422)                   |

## 1.2. Total safety factor

|                               | Gear1<br>[Left] | Gear1<br>[Right] | Gear2<br>[Left] | Gear2<br>[Right] | Required<br>S.F |
|-------------------------------|-----------------|------------------|-----------------|------------------|-----------------|
| Contact S.F                   | 0.4589          |                  | 0.4742          |                  | 1.0000          |
| Bending S.F                   | 0.3869          |                  | 0.3834          |                  | 1.4000          |
| Scuffing S.F<br>(Flash Temp.) | 2.3273          |                  |                 |                  | 2.0000          |
| Micropitting S.F              | 0.1699          |                  |                 |                  | 1.0000          |

## 1.3. Total damage & life

|                      | Gear1<br>[Left] | Gear1<br>[Right] | Gear2<br>[Left] | Gear2<br>[Right] |
|----------------------|-----------------|------------------|-----------------|------------------|
| Contact Damage [%]   | 9999.9000       | 0.0000           | 9999.9000       | 0.0000           |
| Bending Damage [%]   | 9999.9000       | 0.0000           | 9999.9000       | 0.0000           |
| Contact Life [hr]    | 0.0000          | $\infty$         | 0.0000          | $\infty$         |
| Bending Life [hr]    | 0.0000          | $\infty$         | 0.0000          | $\infty$         |
| Total Req. Life [hr] | 10000.0000      |                  |                 |                  |

## 1.4. Others

|                                           | Gear1 - Gear2 |
|-------------------------------------------|---------------|
| Transverse Contact Ratio                  | 1.5100        |
| Overlap Ratio                             | 0.0000        |
| Total Contact Ratio                       | 1.5100        |
| Worst Pitch Line Velocity [m/s]           | 2.9746        |
| Total Efficiency [%]                      | 98.9805       |
| Worst Mesh Loss [W]                       | 1080.4802     |
| Worst Churning Loss [W]                   | 13.1287       |
| Worst Peak-to-Peak T.E. [ $\mu\text{m}$ ] | 22.4853       |
| Gear Total Mass [kg]                      | 1.1570        |
| Hunting tooth                             | Complete      |

## 1.5. Summary - Load cases

[illegible]

## 2. Detail geometry

### 2.1. Input gear geometry

Meshing pair: Gear1 - Gear2

|                                                    | Symbol           | Pair1<br>Gear1                             | Pair1<br>Gear2                             |
|----------------------------------------------------|------------------|--------------------------------------------|--------------------------------------------|
| Normal module (mm)                                 | $m_n$            | 2.0000                                     |                                            |
| Pressure angle at normal section (deg)             | $\alpha_n$       | 20.0000                                    |                                            |
| Helix angle at reference circle (deg)              | $\beta$          | 0.0000                                     |                                            |
| Manual normal backlash (mm)                        | $j_{bn}$         | 0.0000                                     |                                            |
| Number of teeth                                    | $z$              | 28.0000                                    | 41.0000                                    |
| Facewidth (mm)                                     | $b$              | 45.0000                                    | 45.0000                                    |
| Profile shift coefficient                          | $x$              | 0.3132                                     | 0.2129                                     |
| Quality (ISO)                                      | $Q$              | 6.0000                                     | 6.0000                                     |
| Center distance (mm)                               | $a$              | 70.0000                                    |                                            |
| Tooth thickness tolerance                          | -                | DIN 3967 cd25                              | DIN 3967 cd25                              |
| Addendum coefficient                               | $h^*_aP$         | 1.0000                                     | 1.0000                                     |
| Dedendum coefficient                               | $h^*_fP$         | 1.2500                                     | 1.2500                                     |
| Root radius coefficient                            | $\rho^*_fP$      | 0.3800                                     | 0.3800                                     |
| Tip form height coefficient                        | $h^*_{FaP}$      | 0.9500                                     | 0.9500                                     |
| Ramp angle (deg)                                   | $\alpha_{KP}$    | 50.0000                                    | 50.0000                                    |
| Protuberance height coefficient                    | $h^*_{prP}$      | 0.0000                                     | 0.0000                                     |
| Protuberance angle (deg)                           | $\alpha_{prP}$   | 0.0000                                     | 0.0000                                     |
| Tip alteration (mm)                                | $k_{m_n}$        | 0.0000                                     | 0.0000                                     |
| Tip radius coefficient (for topping)               | $\rho^*_{aP}$    | 0.0000                                     | 0.0000                                     |
| Gear material                                      | -                | DIN 18CrNiMo7 ( $\sigma_{Hlim} = 430$ Mpa) | DIN 18CrNiMo7 ( $\sigma_{Hlim} = 430$ Mpa) |
| Young's modulus (N/mm <sup>2</sup> )               | $E$              | $2.0600 \times 10^5$                       | $2.0600 \times 10^5$                       |
| Poisson's ratio                                    | $\nu$            | 0.3000                                     | 0.3000                                     |
| Density (kg/m <sup>3</sup> )                       | $\rho$           | 7830.0000                                  | 7830.0000                                  |
| Coeff. of thermal expansion (10 <sup>-8</sup> /°C) | $\alpha_{coeff}$ | 11.5000                                    | 11.5000                                    |
| Surface hardness                                   | HRC              | 61.0000                                    | 61.0000                                    |
| Core hardness                                      | HBW              | 325.0000                                   | 325.0000                                   |
| Tensile strength (N/mm <sup>2</sup> )              | $\sigma_B$       | 1200.0000                                  | 1200.0000                                  |
| Yield strength (N/mm <sup>2</sup> )                | $\sigma_s$       | 850.0000                                   | 850.0000                                   |
| Specific heat capacity (J/kg.K)                    | $c_M$            | 485.0000                                   | 485.0000                                   |
| Specific heat conductivity (W/m.K)                 | $\lambda_M$      | 50.0000                                    | 50.0000                                    |
| Mean roughness height, root (um)                   | $R_{ZF}$         | 20.0000                                    | 20.0000                                    |
| Mean roughness height, flank (um)                  | $R_{ZH}$         | 4.8000                                     | 4.8000                                     |
| Roughness average value, root (um)                 | $R_{AF}$         | 3.0000                                     | 3.0000                                     |
| Roughness average value, flank (um)                | $R_{AH}$         | 0.6000                                     | 0.6000                                     |

|                                                             | Symbol          | Pair1<br>Gear1 | Pair1<br>Gear2 |
|-------------------------------------------------------------|-----------------|----------------|----------------|
| Fatigue strength for bending stress<br>(N/mm <sup>2</sup> ) | $\sigma_{Flim}$ | 430.0000       | 430.0000       |
| Fatigue strength for contact stress (N/mm <sup>2</sup> )    | $\sigma_{Hlim}$ | 1500.0000      | 1500.0000      |

## 2.2. Calculated geometry - Gear

Meshing pair: Gear1 - Gear2

|                                                                           | Symbol                 | Pair1<br>Gear1    | Pair1<br>Gear2    |
|---------------------------------------------------------------------------|------------------------|-------------------|-------------------|
| Gear ratio                                                                | GR                     | 1.4643            |                   |
| Transverse module (mm)                                                    | $m_t$                  | 2.0000            |                   |
| Working normal pressure angle (deg)                                       | $\alpha_{wn}$          | 22.1396           |                   |
| Transverse pressure angle (deg)                                           | $\alpha_t$             | 20.0000           |                   |
| Working transverse pressure angle (deg)                                   | $\alpha_{wt}$          | 22.1396           |                   |
| Working transverse pressure angle, e/i (deg)                              | $\alpha_{wt}, e/i$     | 22.1587 / 22.1205 |                   |
| Base helix angle (deg)                                                    | $\beta_b$              | 0.0000            |                   |
| Helix angle at operating pitch circle (deg)                               | $\beta_w$              | 0.0000            |                   |
| Radial backlash without tolerance (mm)                                    | $j_{r0}$               | 0.0000            |                   |
| Circumferential backlash without tolerance (mm)                           | $j_{t0}$               | 0.0000            |                   |
| Min/max circumferential backlash by flank tolerance<br>for each gear (mm) | $j_t, e/i$             | 0.0700 / 0.1100   | 0.0700 / 0.1100   |
| Min/max normal backlash by flank tolerance<br>for each gear (mm)          | $j_n, e/i$             | 0.0658 / 0.1034   | 0.0658 / 0.1034   |
| Circumferential backlash, e/i (mm)                                        | $j_{tw}, e/i$          | 0.1343 / 0.2309   |                   |
| Normal backlash, e/i (mm)                                                 | $j_{nw}, e/i$          | 0.1244 / 0.2139   |                   |
| Angular backlash, min (deg)                                               | $j_\theta, \text{min}$ | 0.2709            | 0.1850            |
| Angular backlash, max (deg)                                               | $j_\theta, \text{max}$ | 0.4658            | 0.3181            |
| Effective facewidth (mm)                                                  | $b_{eff}$              | 45.0000           |                   |
| Sum of the profile shift coefficients                                     | Sum_x                  | 0.5261            |                   |
| Generating profile shift coefficient                                      | $x_E, e/i$             | 0.2651 / 0.2376   | 0.1648 / 0.1373   |
| Prevent undercut profile shift coefficient                                | $x_{Emin}$             | -0.6377           | -1.3981           |
| Max. profile shift coeff. (min top land)                                  | $x_{Emax}$             | 1.2297            | 1.6118            |
| Reference center distance, $x_1=x_2=0$ (mm)                               | $a_d$                  | 69.0000           |                   |
| Working center distance (mm)                                              | $a_w$                  | 70.0000           |                   |
| Reference diameter (mm)                                                   | $d$                    | 56.0000           | 82.0000           |
| Operating pitch diameter (mm)                                             | $d_w$                  | 56.8116           | 83.1884           |
| Operating pitch diameter with tolerance (mm)                              | $d_w, e/i$             | 56.8193 / 56.8039 | 83.1997 / 83.1771 |
| 27                                                                        |                        | 61.2528           | 86.8514           |
| Tip diameter with tolerance (mm)                                          | $d_a, e/i$             | 61.2528 / 61.1528 | 86.8514 / 86.7514 |

|                                                             | Symbol    | Pair1<br>Gear1    | Pair1<br>Gear2    |
|-------------------------------------------------------------|-----------|-------------------|-------------------|
| Root diamter (mm)                                           | d_f       | 52.2528           | 77.8514           |
| Root diameter with tolerance (mm)                           | d_f, e/i  | 52.0605 / 51.9506 | 77.6591 / 77.5492 |
| Tip form diameter (mm)                                      | d_Fa      | 61.0528           | 86.6514           |
| Tip form diameter with tolerance (mm)                       | d_Fa, e/i | 60.8313 / 60.7051 | 86.4390 / 86.3180 |
| Root form diamter (mm)                                      | d_Ff      | 53.7851           | 79.3246           |
| Root form diameter with tolerance (mm)                      | d_Ff, e/i | 53.6717 / 53.6094 | 79.1930 / 79.1194 |
| Active tip diameter (EAP) (mm)                              | d_Na      | 61.0528           | 86.6514           |
| Active tip diameter with tolerance (mm)                     | d_Na, e/i | 60.8313 / 60.7051 | 86.4390 / 86.3180 |
| Active root diameter (SAP) (mm)                             | d_Nf      | 54.2348           | 80.0805           |
| Active root diameter with tolerance (mm)                    | d_Nf, e/i | 54.4301 / 54.3370 | 80.2857 / 80.1873 |
| Mean diameter for mass calculation (mm)                     | d_m       | 56.6042           | 82.2028           |
| Gear mass (kg)                                              | m         | 0.4491            | 0.7079            |
| Gear inertia (kg.m^2)                                       | J         | 0.0000            | 0.0000            |
| Nominal tip clearance (mm)                                  | c         | 0.4479            | 0.4479            |
| Effective tip clearance, e/i (mm)                           | c, e/i    | 0.6585 / 0.5345   | 0.6585 / 0.5345   |
| Min. margin of involute contact, (d_Nf_i - d_Ff_e) / 2 (mm) | c_NffFf   | 0.3327            | 0.4972            |

## 2.3. Calculated geometry - Tooth

Meshing pair: Gear1 - Gear2

|                                          | Symbol     | Pair1<br>Gear1  | Pair1<br>Gear2  |
|------------------------------------------|------------|-----------------|-----------------|
| Tooth depth (mm)                         | h          | 4.5000          | 4.5000          |
| Tooth depth, e/i (mm)                    | h, e/i     | 4.6511 / 4.5462 | 4.6511 / 4.5462 |
| Working depth (mm)                       | h_w        | 4.0521          |                 |
| Addendum (mm)                            | h_a        | 2.6264          | 2.4257          |
| Addendum, e/i (mm)                       | h_a, e/i   | 2.6264 / 2.5764 | 2.4257 / 2.3757 |
| Dedendum (mm)                            | h_f        | 1.8736          | 2.0743          |
| Dedendum, e/i (mm)                       | h_f, e/i   | 1.9698 / 2.0247 | 2.1704 / 2.2254 |
| Number of virtual gear teeth             | z_n        | 28.0000         | 41.0000         |
| Tooth thickness coeff.                   | s_nc       | 1.7988          | 1.7257          |
| Transverse tooth thickness (mm)          | s_t        | 3.5976          | 3.4515          |
| Normal tooth thickness (mm)              | s_n        | 3.5976          | 3.4515          |
| Normal tooth thickness at d_a (mm)       | s_an       | 1.2674          | 1.4336          |
| Normal tooth thickness at d_a, e/i (mm)  | s_an, e/i  | 1.2482 / 1.1471 | 1.4097 / 1.3171 |
| Normal tooth thickness at d_Fa (mm)      | s_Fan      | 1.3815          | 1.5336          |
| Normal tooth thickness at d_Fa, e/i (mm) | s_Fan, e/i | 1.4994 / 1.3861 | 1.6237 / 1.5224 |
| Normal space width at d_f (mm)           | e_fn       | 0.0000          | 1.6049          |
| Normal space width at d_f, e/i (mm)      | e_fn, e/i  | 0.0000 / 0.0000 | 1.6413 / 1.6637 |



|                            | Symbol | Pair1<br>Gear1 | Pair1<br>Gear2 |
|----------------------------|--------|----------------|----------------|
| Normal pitch (mm)          | p_n    | 6.2832         |                |
| Transverse pitch (mm)      | p_t    | 6.2832         |                |
| Normal base pitch (mm)     | p_bn   | 5.9043         |                |
| Transverse base pitch (mm) | p_bt   | 5.9043         |                |
| Axial pitch (mm)           | p_x    | 0.0000         |                |
| Lead (mm)                  | p_z    | 0.0000         | 0.0000         |

## 2.4. Calculated geometry - Mesh

Meshing pair: Gear1 - Gear2

|                                                | Symbol    | Pair1<br>Gear1              | Pair1<br>Gear2 |
|------------------------------------------------|-----------|-----------------------------|----------------|
| Length of the path of contact (A to E) (mm)    | g_α       | 8.9157                      |                |
| Length of path of contact, e/i (mm)            | g_α, e/i  | 8.4880 / 8.1776             |                |
| Length of approach POC of pinion (A to C) (mm) | g_f1      | 4.1428                      |                |
| Length of approach POC of pinion, e/i (mm)     | g_f1, e/i | 3.9244 / 3.8945             |                |
| Length of recess POC of pinion (C to E) (mm)   | g_a1      | 4.7730                      |                |
| Length of recess POC of pinion, e/i (mm)       | g_a1, e/i | 4.5636 / 4.5431             |                |
| Length T1 - A (mm)                             | T1A       | 6.5624                      |                |
| T1A tolerance (mm)                             | e/m/i     | 6.9548 / 6.8626 / 6.7705    |                |
| Length T1 - B (mm)                             | T1B       | 9.5738                      |                |
| T1B tolerance (mm)                             | e/m/i     | 9.3542 / 9.2912 / 9.2281    |                |
| Length T1 - C (mm)                             | T1C       | 10.7051                     |                |
| T1C tolerance (mm)                             | e/m/i     | 10.7154 / 10.7051 / 10.6949 |                |
| Length T1 - D (mm)                             | T1D       | 12.4666                     |                |
| T1D tolerance (mm)                             | e/m/i     | 12.8591 / 12.7669 / 12.6747 |                |
| Length T1 - E (mm)                             | T1E       | 15.4781                     |                |
| T1E tolerance (mm)                             | e/m/i     | 15.2585 / 15.1954 / 15.1324 |                |
| Length T2 - A (mm)                             | T2A       | 19.8181                     |                |
| T2A tolerance (mm)                             | e/m/i     | 19.5848 / 19.5179 / 19.4509 |                |
| Length T2 - B (mm)                             | T2B       | 16.8066                     |                |
| T2B tolerance (mm)                             | e/m/i     | 17.1776 / 17.0893 / 17.0011 |                |
| Length T2 - C (mm)                             | T2C       | 15.6754                     |                |
| T2C tolerance (mm)                             | e/m/i     | 15.6903 / 15.6754 / 15.6604 |                |
| Length T2 - D (mm)                             | T2D       | 13.9139                     |                |
| T2D tolerance (mm)                             | e/m/i     | 13.6806 / 13.6136 / 13.5466 |                |
| Length T2 - E (mm)                             | T2E       | 10.9024                     |                |
| T2E tolerance (mm)                             | e/m/i     | 11.2733 / 11.1851 / 11.0968 |                |
| Length T1 - T2 (mm)                            | T1T2      | 26.3805                     |                |

|                                         | Symbol              | Pair1<br>Gear1              | Pair1<br>Gear2 |
|-----------------------------------------|---------------------|-----------------------------|----------------|
| T1T2 tolerance (mm)                     | e/m/i               | 26.4057 / 26.3805 / 26.3553 |                |
| Diameter of single contact point A (mm) | d_A                 | 54.2348                     | 86.6514        |
| Diameter of single contact point B (mm) | d_B                 | 55.9981                     | 84.0672        |
| Diameter of single contact point C (mm) | d_C                 | 56.8116                     | 83.1884        |
| Diameter of single contact point D (mm) | d_D                 | 58.2308                     | 81.9257        |
| Diameter of single contact point E (mm) | d_E                 | 61.0528                     | 80.0805        |
| Transverse contact ratio                | $\epsilon_{\alpha}$ | 1.5100                      |                |
| Overlap ratio                           | $\epsilon_{\beta}$  | 0.0000                      |                |
| Total contact ratio                     | $\epsilon_{\gamma}$ | 1.5100                      |                |
| Total contact ratio with tolerance      | e/m/i               | 1.438 / 1.411 / 1.385       |                |
| Specific sliding at Tip                 | $\zeta_a$           | 0.5190                      | 0.5151         |
| Specific sliding at Root                | $\zeta_f$           | -1.0624                     | -1.0788        |

## 2.5. Calculated geometry - Accuracy

Meshing pair: Gear1 - Gear2

|                                                             | Symbol  | Pair1<br>Gear1 | Pair1<br>Gear2 |
|-------------------------------------------------------------|---------|----------------|----------------|
| Quality Standard                                            | -       | ISO 1328: 2013 | ISO 1328: 2013 |
| Single pitch deviation ( $\mu\text{m}$ )                    | f_pt    | 8.5000         | 8.5000         |
| Base circle pitch deviation ( $\mu\text{m}$ )               | f_pb    | 8.0000         | 8.0000         |
| Sector pitch deviation over k/8 pitches ( $\mu\text{m}$ )   | F_pk/8T | 18.0000        | 17.0000        |
| Profile form deviation ( $\mu\text{m}$ )                    | f_fa    | 8.5000         | 8.5000         |
| Profile slope deviation ( $\mu\text{m}$ )                   | f_Ha    | 7.0000         | 7.0000         |
| Total profile deviation ( $\mu\text{m}$ )                   | F_α     | 11.0000        | 11.0000        |
| Helix form deviation ( $\mu\text{m}$ )                      | f_fb    | 11.0000        | 11.0000        |
| Helix slope deviation ( $\mu\text{m}$ )                     | f_Hβ    | 9.5000         | 9.5000         |
| Total helix deviation ( $\mu\text{m}$ )                     | F_β     | 15.0000        | 15.0000        |
| Cumulative pith deviation, total ( $\mu\text{m}$ )          | F_p     | 25.0000        | 26.0000        |
| Runout ( $\mu\text{m}$ )                                    | F_r     | 22.0000        | 23.0000        |
| Single flank composite, tooth-to-tooth ( $\mu\text{m}$ )    | f_isT   | 8.0000         | 8.0000         |
| Single flank composite, total ( $\mu\text{m}$ )             | F_isT   | 33.0000        | 34.0000        |
| Tooth-to-tooth radial composite deviation ( $\mu\text{m}$ ) | f_idT   | 9.5000         | 9.5000         |
| Total radial composite deviation ( $\mu\text{m}$ )          | F_idT   | 31.0000        | 31.0000        |

## 2.6. Calculated geometry - Measurement

Meshing pair: Gear1 - Gear2

|                                                    | Symbol             | Pair1<br>Gear1    | Pair1<br>Gear2    |
|----------------------------------------------------|--------------------|-------------------|-------------------|
| Number of teeth spanned                            | k                  | 4.0000            | 5.0000            |
| Base tangent length (mm)                           | W_k                | 21.8777           | 28.0088           |
| Base tangent length with tolerance (mm)            | W_k, e/i           | 21.8119 / 21.7743 | 27.9431 / 27.9055 |
| Tolerance of W_k (mm)                              | $\Delta W_k$ , e/i | -0.0658 / -0.1034 | -0.0658 / -0.1034 |
| Diameter of contact point (mm)                     | d_M                | 56.9570           | 81.9586           |
| Theoretical diameter of ball/pin (mm)              | D_M.theory         | 3.6264            | 3.4803            |
| Effective Diameter of ball/pin (mm)                | D_M.eff            | 3.7500            | 3.5000            |
| Radial single ball measurement (mm)                | M_rk               | 31.4005           | 43.8707           |
| Radial single ball measurement with tolerance (mm) | M_rk, e/i          | 31.3276 / 31.2857 | 43.7889 / 43.7418 |
| Diametral two ball measurement (mm)                | M_dk               | 62.8009           | 87.6795           |
| Diametral two ball measurement with tolerance (mm) | M_dk, e/i          | 62.6553 / 62.5714 | 87.5161 / 87.4219 |
| Tolerance of M_dk (mm)                             | $\Delta M_{dk}$    | -0.1457 / -0.2296 | -0.1635 / -0.2576 |

### 3. Detail rating

#### 3.1. General

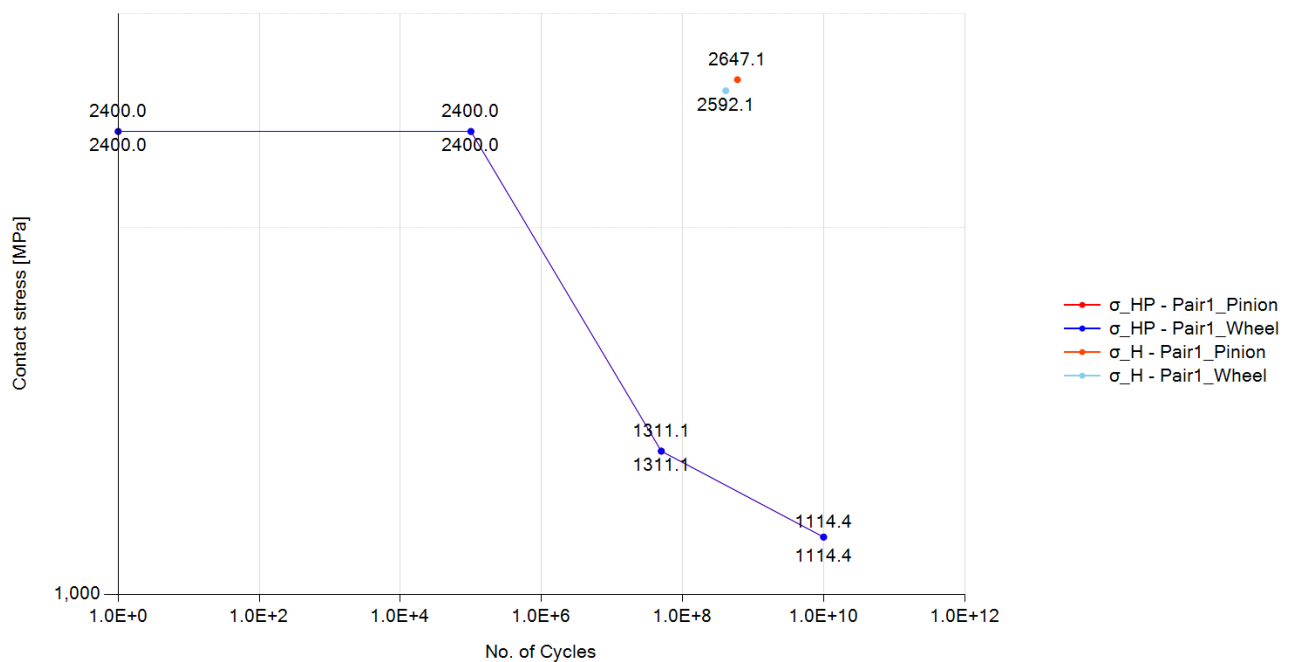
Load case 1, Meshing pair: Gear1 - Gear2

|                                                                               | Symbol        | Pair1<br>Gear1    | Pair1<br>Gear2 |
|-------------------------------------------------------------------------------|---------------|-------------------|----------------|
| Rotational speed (rpm)                                                        | $\omega$      | 1000.0000         | 682.9270       |
| Nominal tangential load at d (N)                                              | $F_t$         | 36585.3660        |                |
| Nominal axial load at d (N)                                                   | $F_a$         | 0.0000            |                |
| Nominal radial load at d (N)                                                  | $F_r$         | 13315.9840        |                |
| Nominal normal load at d (N)                                                  | $F_n$         | 38933.3330        |                |
| Nominal tangential load per unit facewidth (N/mm)                             | $w$           | 813.0080          |                |
| Circumferential speed at d (m/s)                                              | $v$           | 2.9320            |                |
| Sliding velocity at Tip dia. (m/s)                                            | $v_g$         | 0.8410            | 0.7300         |
| Nominal tangential load at $d_w$ (N)                                          | $F_{tw}$      | 36062.7220        |                |
| Nominal axial load at $d_w$ (N)                                               | $F_{aw}$      | 0.0000            |                |
| Nominal radial load at $d_w$ (N)                                              | $F_{rw}$      | 14672.5780        |                |
| Circumferential speed at $d_w$ (m/s)                                          | $v_w$         | 2.9750            |                |
| Running-in value ( $\mu\text{m}$ )                                            | $y_p$         | 0.6000            |                |
| Running-in value ( $\mu\text{m}$ )                                            | $y_f$         | 0.6370            |                |
| Effective single pitch & profile deviation ( $\mu\text{m}$ )                  | $f_{pb\_eff}$ | 7.4000            |                |
| Effective single pitch & profile deviation ( $\mu\text{m}$ )                  | $f_{fa\_eff}$ | 7.8620            |                |
| Correction factor                                                             | $C_M$         | 0.8000            |                |
| Gear blank factor                                                             | $C_R$         | 0.8980            |                |
| Basic rack factor                                                             | $C_B$         | 0.9750            |                |
| Material factor                                                               | $E/E_{st}$    | 1.0000            |                |
| Theoretical single stiffness (N/mm/ $\mu\text{m}$ )                           | $c'_{th}$     | 18.2760           |                |
| Single tooth stiffness (N/mm/ $\mu\text{m}$ )                                 | $c'$          | 12.8010           |                |
| Mesh stiffness (for $K_v$ , $KH\alpha$ , $KF\alpha$ ), (N/mm/ $\mu\text{m}$ ) | $c_{y\alpha}$ | 17.6980           |                |
| Mesh stiffness (for $KH\beta$ , $KF\beta$ ), (N/mm/ $\mu\text{m}$ )           | $c_{y\beta}$  | 15.0430           |                |
| Mean diameter for $K_v$ (mm)                                                  | $d_m (K_v)$   | 56.7530           | 82.3510        |
| Moment of inertia for $K_v$ ( $\text{kg}\cdot\text{mm}^2$ )                   | $J (K_v)$     | 321.4270          | 1203.4950      |
| Moment of gear inertia per unit facewidth ( $\text{kg}\cdot\text{mm}$ )       | $J^*$         | 7.1430            | 26.7440        |
| Relative gear mass per unit facewidth ( $\text{kg}/\text{mm}$ )               | $m^*$         | 0.0103            | 0.0180         |
| Relative gear mass per unit facewidth ( $\text{kg}/\text{mm}$ )               | $m_{red}$     | 0.0066            |                |
| Resonance running speed (rpm)                                                 | $n_{E1}$      | 17713.4280        |                |
| Resonance ratio                                                               | $N$           | 0.0560            |                |
| Resonance ratio in main resonance ratio                                       | $N_s$         | 0.8500            |                |
| Resonance range                                                               | -             | subcritical range |                |
| Dynamic factor                                                                | $K_v$         | 1.0170            |                |

|                                | Symbol        | Pair1<br>Gear1       | Pair1<br>Gear2       |
|--------------------------------|---------------|----------------------|----------------------|
| Face load factor - flank       | $K_{H\beta}$  | 1.3000               |                      |
| Face load factor - root        | $K_{F\beta}$  | 1.2660               |                      |
| Transverse load factor - flank | $K_{H\alpha}$ | 1.0000               |                      |
| Transverse load factor - root  | $K_{F\alpha}$ | 1.0000               |                      |
| Number of load cycles          | $N_L$         | $6.0000 \times 10^8$ | $4.0980 \times 10^8$ |

### 3.2. Pitting

S-N curve (Pitting), Load case 1, Meshing pair: Gear1 - Gear2



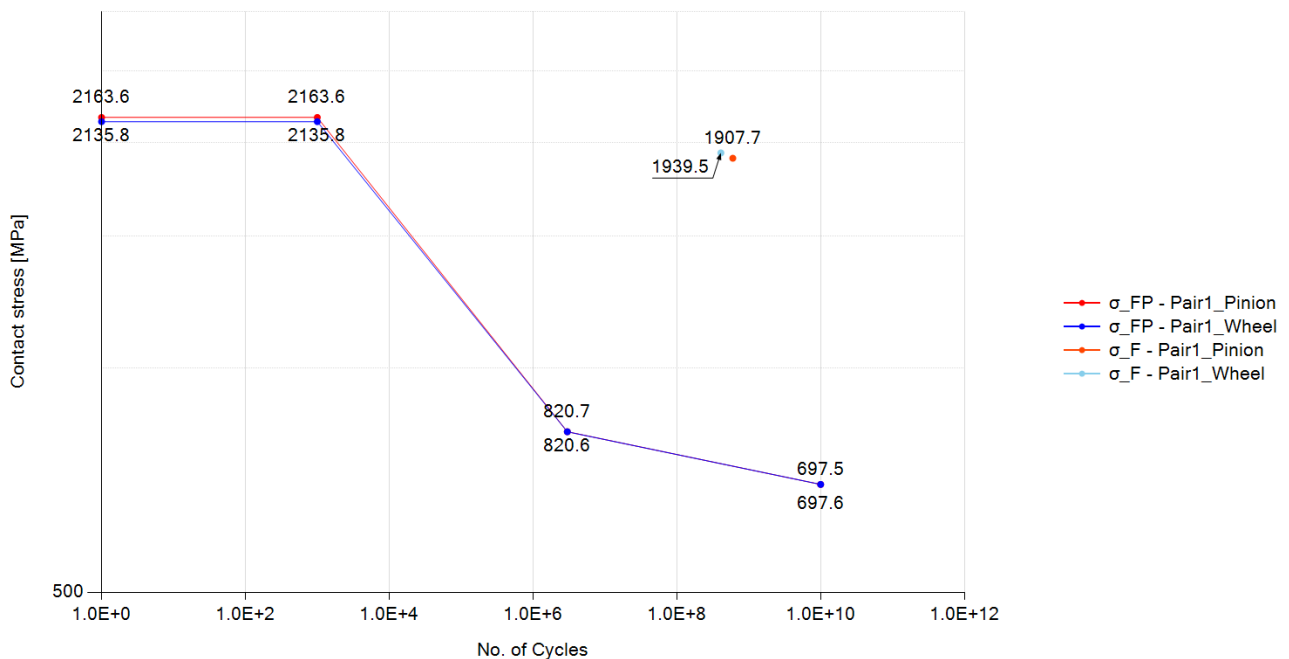
Load case 1, Meshing pair: Gear1 - Gear2

|                                                                | Symbol        | Pair1<br>Gear1       | Pair1<br>Gear2       |
|----------------------------------------------------------------|---------------|----------------------|----------------------|
| Rating Method                                                  | -             | ISO 6336-2:2019      |                      |
| Contact ratio factor                                           | $Z_\epsilon$  | 0.9110               |                      |
| Zone factor                                                    | $Z_H$         | 2.3590               |                      |
| Elasticity factor                                              | $Z_E$         | 189.8120             |                      |
| Young's modulus for tooth flank (N/mm <sup>2</sup> )           | E             | $2.0600 \times 10^5$ | $2.0600 \times 10^5$ |
| Helix angle factor                                             | $Z_\beta$     | 1.0000               |                      |
| Single pair tooth contact factor (pinion)                      | $Z_B$         | 1.0210               |                      |
| Single pair tooth contact factor (wheel)                       | $Z_D$         | 1.0000               |                      |
| Nominal contact stress at the pitch point (N/mm <sup>2</sup> ) | $\sigma_{H0}$ | 2016.7360            |                      |
| Contact stress (N/mm <sup>2</sup> )                            | $\sigma_H$    | 2647.0690            | 2592.0550            |
| Life factor                                                    | $Z_{NT}$      | 0.9270               | 0.9380               |

|                                                             | Symbol                | Pair1<br>Gear1          | Pair1<br>Gear2          |
|-------------------------------------------------------------|-----------------------|-------------------------|-------------------------|
| Lubricant factor                                            | Z_L                   | 0.9460                  | 0.9460                  |
| Velocity factor                                             | Z_V                   | 0.9710                  | 0.9710                  |
| Roughness factor                                            | Z_R                   | 0.9520                  | 0.9520                  |
| Work hardening factor                                       | Z_W                   | 1.0000                  | 1.0000                  |
| Size factor                                                 | Z_X                   | 1.0000                  | 1.0000                  |
| Permissible contact stress - static (N/mm <sup>2</sup> )    | $\sigma_{HP\_static}$ | 2400.0000               | 2400.0000               |
| Permissible contact stress - mid point (N/mm <sup>2</sup> ) | $\sigma_{HP\_mid}$    | 0.0000                  | 0.0000                  |
| Permissible contact stress - reference (N/mm <sup>2</sup> ) | $\sigma_{HP\_ref}$    | 1311.0740               | 1311.0740               |
| Permissible contact stress - long (N/mm <sup>2</sup> )      | $\sigma_{HP\_long}$   | 1114.4130               | 1114.4130               |
| Permissible contact stress (N/mm <sup>2</sup> )             | $\sigma_{HP}$         | 1214.8550               | 1229.1500               |
| Safety factor for Contact                                   | S_H                   | 0.4590                  | 0.4740                  |
| Contact damage (%)                                          | -                     | $6.0000 \times 10^{10}$ | $4.0976 \times 10^{10}$ |
| Cycle to failure                                            | -                     | 1.0000                  | 1.0000                  |

### 3.3. Bending

#### S-N curve (Bending), Load case 1, Meshing pair: Gear1 - Gear2



#### Load case 1, Meshing pair: Gear1 - Gear2

|                          | Symbol | Pair1<br>Gear1  | Pair1<br>Gear2 |
|--------------------------|--------|-----------------|----------------|
| Rating Method            | -      | ISO 6336-3:2019 |                |
| Tooth form factor        | Y_F    | 1.4480          | 1.4940         |
| Stress correction factor | Y_S    | 2.0140          | 1.9850         |

|                                                                | Symbol                | Pair1<br>Gear1 | Pair1<br>Gear2 |
|----------------------------------------------------------------|-----------------------|----------------|----------------|
| Working angle (deg)                                            | $\alpha_{Fen}$        | 22.7250        | 21.8050        |
| Bending moment arm (mm)                                        | $h_{Fe}$              | 2.2740         | 2.3960         |
| Tooth thickness at root (mm)                                   | $s_{Fn}$              | 4.3000         | 4.3600         |
| Tooth root radius (mm)                                         | $\rho_F$              | 0.9620         | 0.9800         |
| Helix angle factor                                             | $Y_{\beta}$           | 1.0000         |                |
| Deep tooth factor                                              | $Y_{DT}$              | 1.0000         |                |
| Rim thickness factor                                           | $Y_B$                 | 1.0000         | 1.0000         |
| Nominal tooth root stress (N/mm <sup>2</sup> )                 | $\sigma_{F0}$         | 1185.9400      | 1205.7160      |
| Tooth root stress (N/mm <sup>2</sup> )                         | $\sigma_F$            | 1907.7370      | 1939.5480      |
| Life factor                                                    | $Y_{NT}$              | 0.8990         | 0.9060         |
| Relative notch sensitive factor                                | $Y_{\delta relT}$     | 0.9970         | 0.9970         |
| Surface factor                                                 | $Y_{RrelT}$           | 0.9570         | 0.9570         |
| Size factor                                                    | $Y_X$                 | 1.0000         | 1.0000         |
| Stress correction factor                                       | $Y_{ST}$              | 2.0000         |                |
| Mean stress influence factor                                   | $Y_M$                 | 1.0000         | 1.0000         |
| Technical factor                                               | $Y_T$                 | 1.0000         | 1.0000         |
| Permissible tooth root stress - static (N/mm <sup>2</sup> )    | $\sigma_{FP\_static}$ | 2163.5850      | 2135.7910      |
| Permissible tooth root stress - reference (N/mm <sup>2</sup> ) | $\sigma_{FP\_ref}$    | 820.6950       | 820.6110       |
| Permissible tooth root stress - long (N/mm <sup>2</sup> )      | $\sigma_{FP\_long}$   | 697.5910       | 697.5190       |
| Permissible tooth root stress (N/mm <sup>2</sup> )             | $\sigma_{FP}$         | 738.0410       | 743.6260       |
| Safety factor for Bending                                      | $S_F$                 | 0.3870         | 0.3830         |
| Bending damage (%)                                             | -                     | $\infty$       | $\infty$       |
| Cycle to failure                                               | -                     | 0.0000         | 0.0000         |

### 3.4. Scuffing

Load case 1, Meshing pair: Gear1 - Gear2

|                                       | Symbol         | Pair1<br>Gear1    | Pair1<br>Gear2 |
|---------------------------------------|----------------|-------------------|----------------|
| Lubrication type                      | -              | Spray lubrication |                |
| Lubricant temperature (deg)           | $\theta_{oil}$ | 80.0000           |                |
| Lubricant viscosity (mPa.s)           | $\eta_{oil}$   | 13.1510           |                |
| Load stage according to FZG A/8, 3/90 | $S_{FZG}$      | 12.0000           |                |
| Gear density (kg/m <sup>3</sup> )     | $\rho_M$       | 7830.0000         | 7830.0000      |
| Gear specific heat (J/kg.K)           | $c_M$          | 485.0000          | 485.0000       |
| Gear heat conductivity (N/s.K)        | $\lambda_M$    | 50.0000           | 50.0000        |
| Gear roughness ( $\mu m$ )            | $R_a$          | 0.6000            | 0.6000         |
| Number of mating gears                | $n_p$          | 1.0000            |                |
| Effective facewidth (mm)              | $b_{eff}$      | 38.2500           |                |

|                                                                       | Symbol    | Pair1<br>Gear1 | Pair1<br>Gear2 |
|-----------------------------------------------------------------------|-----------|----------------|----------------|
| Relevant tip relief ( $\mu\text{m}$ ) (Min. value = 2 $\mu\text{m}$ ) | C_a       | 2.0000         | 2.0000         |
| Transverse unit load (N/mm)                                           | w_Bt      | 1343.0290      |                |
| Lubrication system factor                                             | X_S       | 1.2000         |                |
| Multiple meshing factor                                               | X_mp      | 1.0000         |                |
| Structural factor                                                     | X_W       | 1.0000         | 1.0000         |
| Structural factor (total)                                             | X_Wtot    | 1.0000         |                |
| Thermal contact factor                                                | B_M       | 13.7800        | 13.7800        |
| Multiple path factor                                                  | K_mp      | 1.0000         |                |
| Optimal tip relief ( $\mu\text{m}$ )                                  | C_eff     | 79.3880        |                |
| Angle factor                                                          | X_αβ      | 1.0090         |                |
| Lubricant factor                                                      | X_L       | 0.8790         |                |
| Roughness factor                                                      | X_R       | 0.7750         |                |
| Mean coefficient of friction                                          | μ_m       | 0.1010         |                |
| Maximum flash temperature (deg)                                       | θ_fl_max  | 90.9900        |                |
| Mean flash temperature (deg)                                          | θ_fl_mean | 55.6750        |                |
| Total mass temperature (deg)                                          | θ_M       | 111.4010       |                |
| Scuffing temperature (deg)                                            | θ_S       | 364.8370       |                |
| Maximum contact temperature (deg)                                     | θ_Bmax    | 202.3900       |                |
| Safety factor for flash temperature                                   | S_B       | 2.3270         |                |

### 3.5. Micro-pitting

Load case 1

|                                                                  | Symbol  | Gear1 - Gear2          |
|------------------------------------------------------------------|---------|------------------------|
| minimum lubricant film thickness ( $\mu\text{m}$ )               | h_Ymin  | 0.0215                 |
| minimum specific lubricant film thickness in the contact area    | λ_GFmin | 0.0359                 |
| permissible specific lubricant film thickness                    | λ_GFP   | 0.2111                 |
| safety factor against micropitting                               | S_λ     | 0.1699                 |
| transverse tangential load at reference cylinder per mesh (N)    | F_t     | 36585.3660             |
| transverse load in plane of action (N)                           | F_bt    | 38933.3330             |
| reduced modulus of elasticity (N/mm <sup>2</sup> )               | E_r     | 2.2637×10 <sup>5</sup> |
| thermal contact coefficient of pinion (N/(ms <sup>0.5</sup> *K)) | B_M1    | 13779.6040             |
| thermal contact coefficient of wheel (N/(ms <sup>0.5</sup> *K))  | B_M2    | 13779.6040             |
| failure load stage according to FVA 54/7 (90°)                   | -       | 0.0000                 |
| effective arithmetic mean roughness value ( $\mu\text{m}$ )      | R_a     | 0.6000                 |
| roughness factor                                                 | X_R     | 1.2190                 |
| lubricant factor                                                 | X_L     | 1.0000                 |
| lubrication factor                                               | X_S     | 1.2000                 |



|                                                                                                          | Symbol                | Gear1 - Gear2            |
|----------------------------------------------------------------------------------------------------------|-----------------------|--------------------------|
| elasticity factor (N/mm <sup>2</sup> ) <sup>0.5</sup>                                                    | Z_E                   | 189.8120                 |
| helical load factor                                                                                      | K_B $\gamma$          | 1.0000                   |
| load losses factor                                                                                       | H_v                   | 0.1200                   |
| mean coefficient of friction                                                                             | $\mu_m$               | 0.1220                   |
| profile position at min. lubricant film thickness                                                        | -                     | 0.0000                   |
| tip relief factor                                                                                        | X_Ca                  | 1.0000                   |
| load sharing factor                                                                                      | X_A                   | 0.3330                   |
| buttressing factor                                                                                       | X_but,A               | 0.0000                   |
| effective tip relief ( $\mu\text{m}$ )                                                                   | C_eff                 | 0.0000                   |
| Applied tip relief in calculation ( $\mu\text{m}$ )                                                      | C_a                   | 0.0000                   |
| sum of tangential velocities at point A (m/s)                                                            | $v_{\Sigma,A}$        | 2.1050                   |
| sliding velocity at point A (m/s)                                                                        | $v_{g,A}$             | -0.7300                  |
| tangential velocity on pinion at point A (m/s)                                                           | $v_{r1,A}$            | 0.6870                   |
| tangential velocity on wheel at point A (m/s)                                                            | $v_{r2,A}$            | 1.4170                   |
| oil inlet/sump temperature (°C)                                                                          | $\theta_{oil}$        | 80.0000                  |
| bulk temperature (°C)                                                                                    | $\theta_M$            | 110.9390                 |
| flash temperature at point A (°C)                                                                        | $\theta_{fl,A}$       | 95.0630                  |
| contact temperature at point A (°C)                                                                      | $\theta_{B,A}$        | 206.0020                 |
| pressure-viscosity coefficient at 38 °C (m <sup>2</sup> /N)                                              | $\alpha_{38}$         | $1.8390 \times 10^{-8}$  |
| pressure-viscosity coefficient at bulk temperature (m <sup>2</sup> /N)                                   | $\alpha_{\theta M}$   | $1.2590 \times 10^{-8}$  |
| pressure-viscosity coefficient at point A contact temperature (m <sup>2</sup> /N)                        | $\alpha_{\theta B,A}$ | $7.6890 \times 10^{-9}$  |
| kinematic viscosity at 40 °C (mm <sup>2</sup> /s)                                                        | $\nu_{40}$            | 68.0000                  |
| kinematic viscosity at 100 °C (mm <sup>2</sup> /s)                                                       | $\nu_{100}$           | 8.8000                   |
| kinematic viscosity at point A contact temperature (mm <sup>2</sup> /s)                                  | $\nu_{\theta B,A}$    | 1.7990                   |
| dynamic viscosity at 38 °C (Ns/m <sup>2</sup> )                                                          | $\eta_{38}$           | 0.0650                   |
| dynamic viscosity at bulk temperature (Ns/m <sup>2</sup> )                                               | $\eta_{\theta M}$     | 0.0060                   |
| dynamic viscosity at oil inlet/sump temperature (Ns/m <sup>2</sup> )                                     | $\eta_{\theta oil}$   | 0.0120                   |
| dynamic viscosity at point A contact temperature (Ns/m <sup>2</sup> )                                    | $\eta_{\theta B,A}$   | 0.0010                   |
| normal radius of relative curvature at point A (mm)                                                      | $\rho_{n,A}$          | 4.9300                   |
| material parameter                                                                                       | G_M                   | 2851.0630                |
| local velocity parameter at point A                                                                      | U_A                   | $5.3260 \times 10^{-12}$ |
| local load parameter at point A                                                                          | W_A                   | 0.0004                   |
| local sliding parameter at point A                                                                       | S_GF,A                | 0.1460                   |
| local nominal Hertzian contact stress at point A (Method B)<br>(N/mm <sup>2</sup> )                      | $p_{H,A}$             | 1451.7660                |
| local Hertzian contact stress at point A including the load factors K<br>(Method B) (N/mm <sup>2</sup> ) | $p_{dyn,A}$           | 1865.9150                |

## 4. Power loss

### 4.1. Mesh loss

Load case 1

|                                                     | Symbol       | Gear1 - Gear2       |
|-----------------------------------------------------|--------------|---------------------|
| Rating Method                                       | -            | ISO/TR 14179-2:2001 |
| Input Power (kW)                                    | P            | 107.2739            |
| Gear ratio                                          | u            | 1.4643              |
| Tangential speed at working pitch dia. (m/s)        | $v_{wt}$     | 2.9746              |
| Tangential force at pitch dia. (N)                  | $F_t$        | 36585.3659          |
| Tooth normal force, transverse section (N)          | $F_{bt}$     | 38933.3331          |
| Arithmetic average roughness of pinion              | Ra1          | 0.6000              |
| Arithmetic average roughness of wheel               | Ra2          | 0.6000              |
| Roughness factor                                    | $X_R$        | 0.6000              |
| Sum velocity (m/s)                                  | $v_{\Sigma}$ | 2.2421              |
| Equivalent radius of curvature (mm)                 | $\rho$       | 6.3610              |
| Kinematic viscosity of oil at operating temp. (cSt) | $\eta_{oil}$ | 14.8602             |
| Lubricant type                                      | -            | Mineral oil base    |
| Lubricant factor                                    | $X_L$        | 1.0000              |
| distributed force on tooth surface(N/m)             | $F/b$        | 865.1852            |
| Mean coefficient of friction of the gear mesh       | $\mu_{mz}$   | 0.0839              |
| Tooth loss factor                                   | $H_v$        | 0.1200              |
| Mesh power loss (W)                                 | $P_{VZP}$    | 1080.4802           |
| Mesh efficiency (%)                                 | $\eta_z$     | 98.9928             |

### 4.2. Churning loss

Load case 1

|                                              | Symbol   | Gear1            | Gear2            |
|----------------------------------------------|----------|------------------|------------------|
| Rating Method                                | -        | Changenet (2011) | Changenet (2011) |
| Rotating speed (rpm)                         | n        | 1000.0000        | 682.9268         |
| Lubricant                                    | -        | Oil: ISO-VG 68   | Oil: ISO-VG 68   |
| Gear immersed depth (mm)                     | $h_e$    | 28.6264          | 41.4257          |
| Oil density (kg/m <sup>3</sup> )             | $\rho$   | 885.0000         | 885.0000         |
| Operating Viscosity (cSt)                    | $\nu$    | 14.8602          | 14.8602          |
| dynamic viscosity (kg/m.s)                   | $\mu$    | 0.0000           | 0.0000           |
| Reynolds number                              | Re       | 9077.8568        | 8971.4126        |
| Froude number                                | Fr       | 32.0004          | 21.5977          |
| Centrifugal acceleration (m/s <sup>2</sup> ) | $\gamma$ | 150.3351         | 79.3067          |

|                                  | Symbol | Gear1   | Gear2   |
|----------------------------------|--------|---------|---------|
| Tip diameter of gear (mm)        | d_a    | 61.2528 | 86.8514 |
| Pitch diameter of gear (mm)      | d_p    | 57.2528 | 82.8514 |
| Facewidth (mm)                   | b      | 45.0000 | 45.0000 |
| Dimensionless drag torque        | C_m    | 0.0334  | 0.0350  |
| Immersion area of the gear (m^2) | S_m    | 0.0127  | 0.0201  |
| Total oil churning loss (W)      | P      | 5.0376  | 8.0911  |
| Total oil churning torque (N.m)  | T      | 0.0481  | 0.1131  |

### 4.3. Windage loss

Load case 1

|                      | Symbol | Gear1  | Gear2  |
|----------------------|--------|--------|--------|
| Rating Method        | -      | None   | None   |
| Windage loss (W)     | P_WL   | 0.0000 | 0.0000 |
| Windage torque (N.m) | T_WL   | 0.0000 | 0.0000 |